

Case Study- 1

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Case Study: Binary to Gray Code Converter

Questions:

I. What is the rule for converting binary to Gray code?

Ans-: The conversion from a binary number to its Gray code equivalent follows a simple rule based on the **Exclusive-OR (XOR)** operation.

1. The **Most Significant Bit (MSB)** of the Gray code is the **same** as the MSB of the binary number.

2. Moving from left to right (from MSB to LSB), the next Gray code bit is the **XOR** of the current binary bit and the binary bit to its immediate left (the adjacent, more significant bit).

ii. Derive the logic expressions for each output bit

Ans:- Based on the conversion rule, the logic expressions for each output bit (G_3, G_2, G_1, G_0) in terms of the input bits (B_3, B_2, B_1, B_0) are:

- $G_3 = B_3$
- $G_2 = B_3 \oplus B_2$
- $G_1 = B_2 \oplus B_1$
- $G_0 = B_1 \oplus B_0$

The symbol ' $\oplus$ ' denotes the XOR logical operation.

iv. Why is Gray code used in digital communication or error correction systems?

Ans:- Gray code is widely used because of its unique **single-bit change** property. This means that any two successive values in the Gray code sequence differ by only **one** bit. For example, the Gray code for 7 is 0100 and for 8 is 1100, a single

bit change. In contrast, their binary equivalents 0111 and 1000 differ by all four bits.

This property is crucial in systems like:

- **Electromechanical Switches & Encoders:** In position encoders (like shaft encoders), a small misalignment during the transition between two adjacent sectors can cause a standard binary encoder to output a completely incorrect value. For instance, when moving from 3 (011) to 4 (100), a brief misread might yield 111 (7). With Gray code, this ambiguity is eliminated. A misread will only result in either the previous or the next value, making the system much more reliable and error-resistant. □
- **Digital Communication:** It minimizes the impact of single-bit errors during data transmission. A single-bit flip error will result in a value that is numerically adjacent to the original, preventing large, catastrophic errors.
- **State Machines & Karnaugh Maps:** It's used to order the rows and columns in Karnaugh maps, which simplifies the process of finding logic expressions for digital circuits.

v. How would you implement the reverse conversion (Gray to Binary)?

Ans:- The process to convert a Gray code back to its binary equivalent also follows a simple, iterative rule.

Rule for Gray to Binary Conversion:

1. The **MSB** of the binary number is the **same** as the MSB of the Gray code.
2. Moving from left to right, the next binary bit is the **XOR** of the *previously calculated* binary bit and the *current* Gray code bit.

For a 4-bit Gray code $G_3 G_2 G_1 G_0$ and its binary equivalent $B_3 B_2 B_1 B_0$:

- $B_3 = G_3$
- $B_2 = B_3 \oplus G_2$
- $B_1 = B_2 \oplus G_1$
- $B_0 = B_1 \oplus G_0$

Implementation: This can be implemented with a similar circuit, where the XOR gates are cascaded, feeding the output of one stage into the next.

iii. Draw the logic circuit diagram?

Ans:- The logic circuit can be implemented using three **XOR gates**. The input binary bits are B3 ,B2 ,B1 ,B0 and the output Gray code bits are G3 ,G2 ,G1 ,G0 .

- G3 is connected directly to the B3 input.
- An XOR gate takes B3 and B2 as inputs to produce G2 .
- A second XOR gate takes B2 and B1 as inputs to produce G1 .
- A third XOR gate takes B1 and B0 as inputs to produce G0 .

